STAT3003: Survey Methods
6. Using Auxiliary Data

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6.1 Introduction

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- How might you do this?

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- Any problems with this method?

6.1 Introduction

- You are instructed to estimate the total sugar content of a shipping container of oranges
- How might you do this?
  - Take a simple random sample of n oranges, estimate the mean sugar content with $\bar{y}$, then estimate τ with $N\bar{y}$
- Any problems with this method?
  - Finding out what N is
  - You'll destroy lots of oranges
- Is there another way?

6.1 Introduction

- So far, the survey methods we have described depend entirely on $Y_1, Y_2, \dots, Y_n$ for information to estimate a parameter
- Sometimes other variables are closely related to the response variable Y, and can provide useful additional information
- These auxiliary variables can help us estimate parameters in ways that help us save time and money

6.1 Introduction

- Can we think of a variable that is very closely related to the sugar content of an orange but is also a lot easier to measure?
  - Try the weight of an orange
  - Thus when we sample orange i, we record its weight X_i and its sugar content Y_i
  - Our sample will be n pairs of observations $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$

6.1 Introduction

- Some examples of variables of interest (Y) and their auxiliary variable (X)
- Say Hong Kong has N bookstores. Then
  - Y_i = sales of "Idiot's Guide to Statistics" in store i
  - X_i = size of store i
- Say a forest has N trees. Then
  - Y_i = the volume of a tree i
  - X_i = the diameter of a tree i
- Say a population has N elements. Then
  - Y_i = measurement of interest of element i that is difficult or expensive to measure
  - X_i = auxiliary measurement of element i that is (we think) related to Y_i but easy or cheap to measure

6.1 Introduction

- Notation
  - τ_y and τ_x are the population totals for variables Y and X respectively
  - μ_y and μ_x are the population means for variables Y and X respectively
- Since $\frac{\tau_y}{\tau_x} = \frac{\mu_y}{\mu_x}$, then $\tau_y = \frac{\mu_y}{\mu_x}\tau_x$ and we have an alternate way to estimate τ_y
  - $\hat{\tau}_r \coloneqq \frac{\bar{Y}}{\bar{X}}\tau_x$, the ratio estimator for τ_y

6.1 Introduction

- Compare $\hat{\tau}_r$ with $\hat{\tau}$, our previous estimator
  - $\hat{\tau} = N\bar{Y}$
  - $\hat{\tau}_r := \frac{\bar{Y}}{\bar{X}}\tau_x$
  - We see the ratio estimator does not require the population size N
  - Thus ratio estimation may be a good choice if we wish to estimate the population total, but we do not know the population size

6.1 Introduction

- In addition, we shall see that the ratio estimator for τ and μ can be very useful when
  – Y and X are highly linearly correlated through the origin
  – Then the ratio estimator has a lower variance than the ordinary estimator
- Thus we should check whether this kind of relationship exists
  – A scatterplot of your sample $\{(x_1, y_1), \dots, (x_n, y_n)\}$ can provide a quick visual check

6.1 Introduction

- Similarly, we can estimate μ_y by
  - $\hat{\mu}_r = \frac{\bar{Y}}{\bar{X}}\mu_x$
- Or indeed we can estimate a ratio of interest, $R = \frac{\tau_y}{\tau_x} = \frac{\mu_y}{\mu_x}$, by
  - $\hat{R} = \frac{\bar{Y}}{\bar{X}}$
  - Thus $\hat{\tau}_r = \hat{R}\tau_x$ and $\hat{\mu}_r = \hat{R}\mu_x$
- Thus ratio estimators presume population parameters of variable X are known

6.1 Introduction

- Sometimes we want to estimate R, a population ratio
  - When measuring inflation, in fact this is a ratio of the price of a basket of goods measured at two different points in time
  - When forecasting sales over the coming year, we may use the ratio of January sales this year to January sales last year, then multiply this by total sales last year
  - When researching sociology, researchers may use the ratio of family food budget to total income

6.2 Properties of Ratio Estimators

- Are ratio estimators unbiased?
  - No. They are biased.
  - However, when the sampling is SRS and the sample size is large, this bias is very small
- This means we have to consider MSE
  - Recall $MSE(\hat{\mu}_r) = Var(\hat{\mu}_r) + \left(Bias(\hat{\mu}_r)\right)^2$
  - However, since the bias-squared is small relative to the variance for large SRS sample, the first order approximation of MSE is the same as for $Var(\hat{\mu})$

6.2 Properties of Ratio Estimators

- To investigate the bias of the ratio estimator, consider
  - $Cov(\hat{R},\bar{X}) = E[\hat{R}\bar{X}] - E[\hat{R}]E[\bar{X}]$
  - $\Rightarrow Cov(\hat{R},\bar{X}) = \mu_y - E[\hat{R}]\mu_x$
  - $\Rightarrow E[\hat{R}] = \frac{\mu_y - Cov(\hat{R},\bar{X})}{\mu_x} = R - \frac{Cov(\hat{R},\bar{X})}{\mu_x}$
  - Therefore the bias of $\hat{R} = E[\hat{R}] - R = -\frac{Cov(\hat{R},\bar{X})}{\mu_x}$
- Now, it can be shown (by the Cauchy-Schwarz inequality) that for any two random variables X and Y
  - $[Cov(X,Y)]^2 \leq Var(X)Var(Y)$
  - $\Rightarrow \frac{|Bias(\hat{R})|}{\sqrt{Var(\hat{R})}} \leq \frac{\sqrt{Var(\bar{X})}}{\mu_x}$, providing an upper bound for the bias

6.2 Properties of Ratio Estimators

- To work out the variance of $\hat{R}$ (and how to estimate it), we need some calculus.
- The Taylor expansion of a function $f(x, y)$ about a point (a, b) is
  - $f(x,y) = f(a,b) + \frac{\partial f}{\partial x}(a,b)(x-a) + \frac{\partial f}{\partial y}(a,b)(y-b) + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(a,b)(x-a)^2 + \frac{1}{2}\frac{\partial^2 f}{\partial y^2}(a,b)(y-b)^2 + \frac{\partial^2 f}{\partial x \partial y}(a,b)(x-a)(y-b) + \cdots$
- Let's take $f(x,y) = \frac{y}{x}, (a,b) = (\mu_x, \mu_y)$ and evaluate f at $x = \bar{X}, y = \bar{Y}$

6.2 Properties of Ratio Estimators

- Working out the partial derivatives and plugging in the numbers yields
  - $f(\bar{X}, \bar{Y}) = \frac{\bar{Y}}{\bar{X}} = \hat{R} = \frac{\mu_y}{\mu_x} - \frac{\mu_y}{\mu_x^2}(\bar{X} - \mu_x) + \frac{1}{\mu_x}(\bar{Y} - \mu_y) + \frac{\mu_y}{\mu_x^3}(\bar{X} - \mu_x)^2 - \frac{1}{\mu_x^2}(\bar{X} - \mu_x)(\bar{Y} - \mu_y) + \cdots$
- Taking the first-order approximation (i.e. ignoring all quadratic and higher power terms) we have
  - $\hat{R} \approx \frac{\mu_y}{\mu_x} - \frac{\mu_y}{\mu_x^2}(\bar{X} - \mu_x) + \frac{1}{\mu_x}(\bar{Y} - \mu_y)$

6.2 Properties of Ratio Estimators

- Hence
  - $E\left[\left(\hat{R}-R\right)^2\right] \approx \frac{\mu_y^2}{\mu_x^4} Var(\bar{X}) + \frac{1}{\mu_x^2} Var(\bar{Y}) - \frac{2\mu_y}{\mu_x^3} Cov(\bar{X}, \bar{Y})$
  - $= \frac{1}{\mu_x^2}[Var(R\bar{X}) + Var(\bar{Y}) - 2Cov(R\bar{X}, \bar{Y})]$
  - $= \frac{1}{\mu_x^2} Var(\bar{Y} - R\bar{X})$
- In other words, the MSE of $\hat{R}$ is well approximated by $\frac{1}{\mu_x^2} Var(\bar{Y} - R\bar{X})$
  - Recall $MSE(\hat{R}) = Var(\hat{R}) + \left(Bias(\hat{R})\right)^2$, hence $Var(\hat{R}) \approx MSE(\hat{R}) \approx \frac{1}{\mu_x^2} Var(\bar{Y} - R\bar{X})$
  - $\bar{Y} - R\bar{X}$ is the sample mean of $y_i - Rx_i$
  - (x_i, y_i) is a SRS, thus $y_i - Rx_i$ is also a SRS

6.2 Properties of Ratio Estimators

- Thus according to formula for SRS, $Var(\hat{\mu}_r)$ is
  - $Var(\hat{\mu}_r) \approx \frac{N-n}{N-1}\frac{1}{n}\ \sigma_r^2$
  - where $\sigma_r^2 = \frac{1}{N}\sum_{i=1}^{N}(y_i - Rx_i)^2$ is population variance of $y_i - Rx_i$ because E($y_i - Rx_i$)=0
- And a formula for estimating the variance is
  - $\widehat{Var}(\hat{\mu}_r) = \left(1 - \frac{n}{N}\right)\frac{1}{n}\hat{\sigma}_r^2,$
  - where $\hat{\sigma}_r^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(Y_i - \hat{R}X_i\right)^2$

6.2 Properties of Ratio Estimators

- Let's compare $\widehat{Var}(\hat{\mu}_r)$ with $\widehat{Var}(\bar{Y})$
  - $\widehat{Var}(\hat{\mu}_r) = \left(1 - \frac{n}{N}\right)\frac{1}{n}\hat{\sigma}_r^2, \widehat{Var}(\bar{Y}) = \left(1 - \frac{n}{N}\right)\frac{\hat{\sigma}_y^2}{n}$
  - Hence we should prefer to use $\hat{\mu}_r$ when $\hat{\sigma}_r^2 \ll \hat{\sigma}_y^2$
- It can be shown that (Exercise!)
  - $\hat{\sigma}_r^2 = \hat{\sigma}_y^2 + \hat{R}^2\hat{\sigma}_x^2 - 2\hat{R}\hat{\rho}\hat{\sigma}_x\hat{\sigma}_y$ where $\hat{\rho} = \frac{\hat{\sigma}_{xy}}{\hat{\sigma}_x\hat{\sigma}_y}$ is the sample correlation between Y and X
  - $\Rightarrow \frac{\widehat{Var}(\bar{Y})}{\widehat{Var}(\hat{\mu}_r)} = \frac{\hat{\sigma}_y^2}{\hat{\sigma}_r^2} = \frac{\hat{\sigma}_y^2}{\hat{\sigma}_y^2 + \hat{R}^2\hat{\sigma}_x^2 - 2\hat{R}\hat{\rho}\hat{\sigma}_x\hat{\sigma}_y}$
  - Therefore if $\hat{\rho} \gg \frac{1}{2}\frac{\left(\frac{\hat{\sigma}_x}{\bar{X}}\right)}{\frac{\hat{\sigma}_y}{\bar{Y}}}$, then $\hat{\sigma}_r^2 \ll \hat{\sigma}_y^2$

6.2 Properties of Ratio Estimators

- The ratio of standard deviation to mean is called the coefficient of variation (CV)
  - Usually the notation is $\hat{c}_{v,X} := \frac{\widehat{\sigma}_x}{\bar{X}}$
  - The CV shows how variable the population relative to the mean
  - It is dimensionless
- If the CVs of Y and X are similar, we see that if the correlation between X and Y is greater than 0.5, then the ratio estimator is good choice.

6.2 Properties of Ratio Estimators

- We've found the formulae for ratio estimation of μ_y
  - $\hat{\mu}_r = \frac{\bar{Y}}{\bar{X}}\mu_x, Var(\hat{\mu}_r) \approx \frac{N-n}{N-1}\frac{1}{n}\sigma_r^2, \widehat{Var}(\hat{\mu}_r) = \left(1 - \frac{n}{N}\right)\frac{1}{n}\hat{\sigma}_r^2$
- For estimation of τ_y we will use
  - $\hat{\tau}_r = \frac{\bar{Y}}{\bar{X}}\tau_x, Var(\hat{\tau}_r) \approx \frac{N-n}{N-1}\frac{N^2}{n}\sigma_r^2, \widehat{Var}(\hat{\tau}_r) = (N-n)\frac{N}{n}\hat{\sigma}_r^2$
- For estimation of R we will use
  - $\hat{R} = \frac{\bar{Y}}{\bar{X}}, Var(\hat{R}) \approx \frac{1}{\mu_x^2}\frac{N-n}{N-1}\frac{1}{n}\sigma_r^2, \widehat{Var}(\hat{R}) = \frac{1}{\mu_x^2}\left(1 - \frac{n}{N}\right)\frac{1}{n}\hat{\sigma}_r^2$
- where
  - $\sigma_r^2 = \frac{1}{N}\sum_{i=1}^{N}(y_i - Rx_i)^2$ and $\hat{\sigma}_r^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(Y_i - \hat{R}X_i\right)^2$

6.2 Properties of Ratio Estimators

- Confidence intervals will be in the familiar form of
  - $Point\ Estimate\ \pm t_{n-1,1-\frac{\alpha}{2}}\sqrt{\widehat{Var}(Estimator)}$
- To find required sample sizes to achieve a $100(1-\alpha)\%$ C.I. with width $2d$ we can either solve
  - $d = z_{1-\frac{\alpha}{2}}\sqrt{Var(Estimator)}$ for n or
  - $d = t_{n-1,1-\frac{\alpha}{2}}\sqrt{Var(Estimator)}$ for n
  - Note both methods require some reasonable estimate of σ_r^2

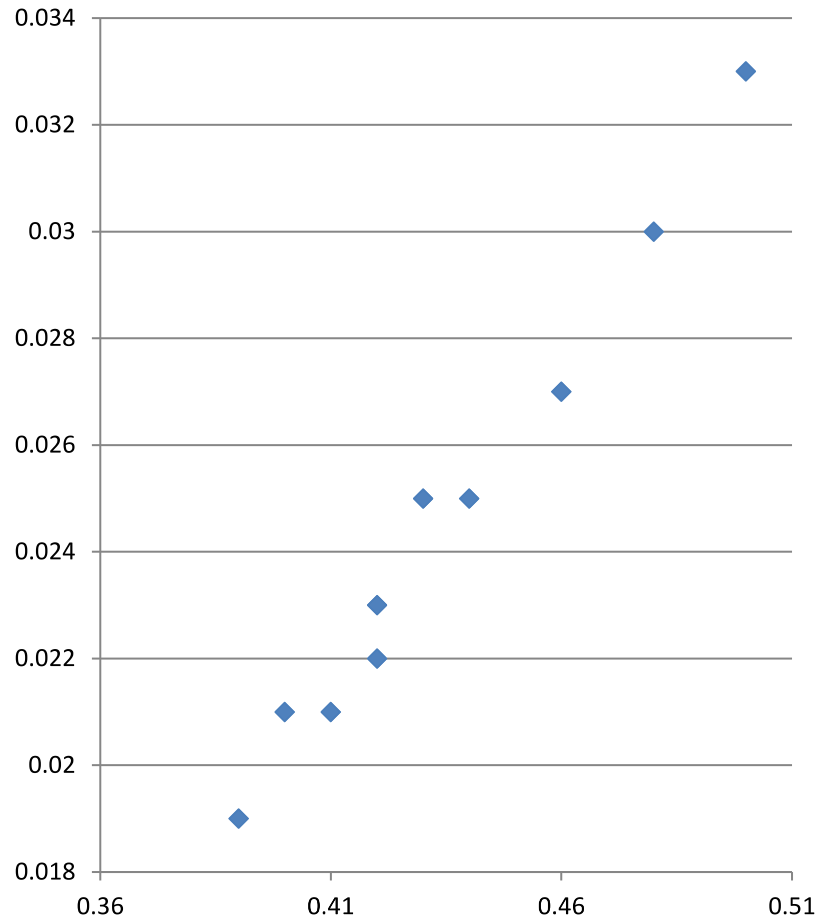
6.2 Properties of Ratio Estimators

- Example from (Scheaffer et al. 2012)
  - We wish to estimate the sugar content of truck load of oranges. We don't know how many oranges are in the load, but by weighing the truck, we find that the total weight of all the oranges is 1800 kgs.
  - We sample 10 oranges, which we weigh and juice, allowing us to find their sugar content. We record the following data

Weight x	**0.4**	**0.48**	**0.43**	**0.42**	**0.5**	**0.46**	**0.39**	**0.41**	**0.42**	**0.44**
Sugar y	0.021	0.03	0.025	0.022	0.033	0.027	0.019	0.021	0.023	0.025

6.2 Properties of Ratio Estimators

- Ratio estimation works best if there is a strict proportional relationship. Before using ratio or regression estimation, we should always check graphically if the desired relationship between y and x holds
- Plotting the data we will a strong, positive association between sugar content and weight



6.2 Properties of Ratio Estimators

- We don't know N, the total number of oranges, so we estimate τ_y using
  - $\hat{\tau}_r = \hat{R}\tau_x = \frac{\bar{y}}{\bar{x}}\tau_x = \frac{0.0249}{0.435} \times 1800 = 101.79\ kg$
- To estimate the variance of the estimate, we use
  - $\widehat{Var}(\hat{\tau}_r) = (N-n)\frac{N}{n}\hat{\sigma}_r^2$
- But this uses N, which we don't know. What can we do?
  - Because N is clearly large, we take $\left(1 - \frac{n}{N}\right) \approx 1$
  - Remember $N = \frac{\tau_x}{\mu_x}$. We know $\tau_x = 1800$, and we replace μ_x by $\bar{x}$
- Hence the estimated variance becomes
  - $\widehat{Var}(\hat{\tau}_r) \approx \frac{\tau_x^2}{\bar{x}^2}\frac{s_r^2}{n} = \frac{1800^2}{0.435^2} \times \frac{0.0024^2}{10} = 9.86$

6.2 Properties of Ratio Estimators

- Therefore our approximate 95% C.I. for τ_y is
  - $101.79 \pm t_{9,0.975}\sqrt{9.86} = (101.79 \pm 7.10)$
- Note that to calculate the standard error, we have had to estimate N using $\widehat{N} = \frac{\tau_x}{\bar{x}}$
- How does the ratio estimate compare with the unbiased estimate?
  - We calculated $\hat{s}_r^2 = 0.0024^2$
  - We calculate $\hat{s}_y^2 = 0.0044^2$ (and we still have to estimate N to calculate $\hat{\tau}_y = N\bar{Y}$)

6.2 Properties of Ratio Estimators

- So ratio estimation reduces the standard error by roughly half
  - The sample correlation is very high: 0.991
  - We can see that the linear model fits well
  - Of course the fit will go through zero (how much sugar can an orange of weight zero have?)
  - The sample CV of X is 0.081, for Y 0.178
  - Hence $\frac{1}{2}\frac{\hat{c}_{v,X}}{\hat{c}_{v,Y}} \approx 0.228 \ll 0.991 = \hat{\rho}$
- In summary, the conditions for this survey were perfect for ratio estimation

6.2 Properties of Ratio Estimators

- Let's do a small population example (from Thompson, 1992) to see bias and MSE in action
- Along a river, you place nets at N sites
  - Site i has x_i nets
  - In total, y_i fish are caught at site i
- We wish to estimate τ_y, the total number of fish caught along the river
  - We conduct an SRS of size n of the N sites, and at each we measure # nets and total # fish caught

6.2 Properties of Ratio Estimators

- Let's make $N = 4$, $n = 2$. The table below completely describes the population

Site i	**1**	**2**	**3**	**4**
# Nets, x_i	4	5	8	5
# Fish, y_i	200	300	500	400

- There are $\frac{4!}{2!2!} = 6$ possible samples
  - $\{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{2,4\}, \{3,4\}$
  - For each we will work out $\hat{t}_r$, $\hat{t}$
  - This will allow us to calculate the bias and MSE of the estimators

6.2 Properties of Ratio Estimators

- For $\hat{\tau}_r$
  - Sample $\{1,2\}$ gives an estimate $\frac{\bar{y}}{\bar{x}}\tau_x = 1222$
  - Sample $\{1,3\}$ gives an estimate…
  - …
  - Sample $\{3,4\}$ gives an estimate 1523
  - Each sample is equally likely (because they come from SRS)
  - Hence $E(\hat{\tau}_r) = \frac{1}{6} \times 1222 + \frac{1}{6} \times \cdots + \frac{1}{6} \times 1523 = 1398.17$
- We know $\tau_y = 1400$, so we see $E(\hat{\tau}_r) \neq \tau_y$ but only slightly
  - The ratio estimator is slightly biased.

6.2 Properties of Ratio Estimators

- Recall for an estimator $\hat{\theta}$ of a parameter θ, the mean squared error is defined by $E\left[\left(\hat{\theta}-\theta\right)^2\right]$
- For $\hat{\tau}_r$
  - $MSE = \frac{1}{6} \times (1222 - 1400)^2 + \cdots + \frac{1}{6} \times (1523 - 1400)^2 = 14451.2$
- For $\hat{\tau}$ from SRS of Y :
  - The possible values are 1000, 1400, 1200, 1600, 1400, and 1800
  - $\Rightarrow E[\hat{\tau}] = \frac{1}{6} \times 1000 + \cdots + \frac{1}{6} \times 1800 = 1400 = \tau_y$
  - $MSE = \frac{1}{6} \times (1000 - 1400)^2 + \cdots + \frac{1}{6} \times (1800 - 1400)^2 = 66667$
- So although $\hat{\tau}_r$ is biased and $\hat{\tau}$ is not, the MSE of $\hat{\tau}_r$ is much smaller

6.2 Properties of Ratio Estimators

- Example
  - We wish to estimate the average # trees per acre on a 1000-acre plantation
  - We sample 10 one-acre plots by SRS and for each count # trees in them
  - We also have aerial photographs of the plantation from which software can estimate # trees on each plot of the whole plantation
  - The software suggests there are 19.7 trees per acre across the plantation

6.2 Properties of Ratio Estimators

- The data is summarized here
- Use a ratio estimator to estimate the mean number of trees per acre across the whole plantation
- Provide a 95% confidence interval

Plot	Actual # per acre	Aerial estimate
1	25	23
2	15	14
3	22	20
4	24	25
5	13	12
6	18	18
7	35	30
8	30	27
9	10	8
10	29	31
mean	22.10	20.80

6.2 Properties of Ratio Estimators

- Ratio estimation isn't a survey design by itself, rather another approach to estimating
  - So far our examples have considered using the SRS design with ratio estimators
  - But ratio estimation does require pairs (x_i, y_i) of observations to made about a sampling unit
- Which designs have we seen which involve measuring two properties of a sampling unit?

6.3 Ratio Estimation and Cluster Sampling

- Recall how we perform cluster sampling
  - Construct a frame of clusters
  - Decide how many you want to sample
  - Choose which to sample through a SRS
  - Visit each cluster selected and measure each element within
  - Then use your data to estimate parameters…
- We can emphasise a key aspect of cluster sampling

6.3 Ratio Estimation and Cluster Sampling

- Recall how we perform cluster sampling
  - Construct a frame of clusters
  - Decide how many you want to sample
  - Choose which to sample through a SRS
  - Visit each cluster selected and observe pair (M_i, Y_i) where Y_i is the cluster total and M_i the cluster population
  - Then use your data to estimate parameters…

6.3 Ratio Estimation and Cluster Sampling

- Here the auxiliary variable is M_i, the population size of cluster i
- We suspect a ratio estimator under cluster sampling would work well if Y_i and M_i are highly correlated
  - That is, if the cluster total and the cluster population size are highly correlated
- If Y_i and M_i are not highly correlated, using the ratio estimator could be a bad choice

6.3 Ratio Estimation and Cluster Sampling

- The (biased) ratio estimator for τ under cluster sampling is
  - $\hat{\tau}_r = \frac{\bar{Y}}{\bar{M}} M = \hat{R} M$, where $M = \sum_{j=1}^{N} M_j$ and $\bar{M} = \frac{1}{n} \sum_{i=1}^{n} M_i$
  - (We've changed notation here. Now $\bar{M}$ means the average cluster size in the sample, not in the population)
  - $\widehat{Var}(\hat{\tau}_r) = N(N-n)\frac{1}{n}\hat{\sigma}_r^2$ where $\hat{\sigma}_r^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(Y_i - \hat{R} M_i\right)^2$

6.3 Ratio Estimation and Cluster Sampling

- Recall the sociologist who wants to estimate τ the total per-capita income in a small city
- She partitioned the city into 415 clusters
  - Used SRS to select 25
  - Visited each cluster and interviewed all the households inside
  - The unbiased estimator gave a C.I. of approximately $(\$18.5M, \$25.7M)$
  - She knows there are 2500 residents in the city
- Now she tries the ratio estimator, for fun (!)

6.3 Ratio Estimation and Cluster Sampling

- The data she recorded

Cluster	No. residents, m_i	Total income per cluster, y_i ($000's)	Cluster	No. residents, m_i	Total income per cluster, y_i ($000's)
1	8	96	14	10	49
2	12	121	15	9	53
3	4	42	16	3	50
4	5	65	17	6	32
5	6	52	18	5	22
6	6	40	19	5	45
7	7	75	20	4	37
8	5	65	21	6	51
9	8	45	22	8	30
10	3	50	23	7	39
11	2	85	24	3	47
12	6	43	25	8	41
13	5	54			

6.3 Ratio Estimation and Cluster Sampling

- The point estimate, $\hat{\tau}_r$ is given by
  - $\hat{\tau}_r = \frac{\bar{y}}{\bar{x}} M = \frac{53160}{6.04} \times 2500 = \$22003311$
  - Recall $\hat{\tau} = 22031400$. The estimates are very close.
- The estimated standard error is given by
  - $s_r^2 = 25189.31^2$
  - $\widehat{Var}(\hat{\tau}_r) = 415 \times (415 - 25) \times 25189.31^2/25 = 2026761^2$
  - $t_{24,0.975} = 2.06$
  - $\Rightarrow$ the 95% C.I. is $(\$17.8M, \$26.2M)$
  - Thus the C.I. using $\hat{\tau}_r$ is wider than that using $\hat{\tau}$

6.3 Ratio Estimation and Cluster Sampling

- Thus the sociologist is not impressed – the ratio has worsened her estimate, not improved it
- She digs a little deeper to find
  - $\hat{\rho} = 0.303$, so the correlation between the total income of a cluster and the size of a cluster is (perhaps surprisingly) quite weak
  - $\hat{c}_{v,Y} = 0.41$, $\hat{c}_{v,M} = 0.39 \Rightarrow \frac{1}{2}\frac{\hat{c}_{v,M}}{\hat{c}_{v,Y}} = 0.48 \gg \hat{\rho}$
  - It seems the cluster sizes provide little information about the cluster totals, hence the unbiased estimator is preferred

6.3 Ratio Estimation and Cluster Sampling

- We didn't derive the ratio formulae for estimating p under SRS. Why?
  - The pairs of observations (x_i, y_i) would look like $(x_1, 1), (x_2, 1), (x_3, 0), \ldots, (x_n, 1)$
  - Hence correlation between Y and auxiliary variable X is not meaningful when Y is binary
- However, estimating p with ratio estimation and cluster sampling works nicely
  - Because p is a ratio itself

6.3 Ratio Estimation and Cluster Sampling

- In cluster i, consider A_i, the number of people in the cluster who have a certain characteristic
- Then $p = \frac{\sum_{i=1}^{N} A_i}{\sum_{i=1}^{N} M_i}$,
  - hence we can try $\hat{p}_r = \frac{\sum_{i=1}^{n} A_i}{\sum_{i=1}^{n} M_i}$ as an estimator of p
  - And $\widehat{Var}(\hat{p}_r) = \frac{N-n}{N} \frac{N^2}{M^2} \frac{\hat{\sigma}_r^2}{n}$
  - where $\hat{\sigma}_r^2 = \frac{1}{n-1} \sum_{i=1}^{n} (A_i - \hat{p}_r M_i)^2$

6.3 Ratio Estimation and Cluster Sampling

- Example
  - When conducting her survey, the sociologist also asked each household whether their house was rented or not.

Cluster	No. residents, m_i	# rented houses, a_i	Cluster	No. residents, m_i	# rented houses, a_i
1	8	4	14	10	5
2	12	7	15	9	4
3	4	1	16	3	1
4	5	3	17	6	4
5	6	3	18	5	2
6	6	4	19	5	3
7	7	4	20	4	1
8	5	2	21	6	3
9	8	3	22	8	3
10	3	2	23	7	4
11	2	1	24	3	0
12	6	3	25	8	3
13	5	2			

6.3 Ratio Estimation and Cluster Sampling

- The point estimate is
  - $\hat{p}_r = \frac{72}{151} = 0.476821$
  - $s_r^2 = 0.530543 = 0.728384^2$
  - $M = 2500 \Rightarrow \frac{M^2}{N^2} = 36.28974$
  - $\widehat{Var}(\hat{p}_r) = \left(1 - \frac{25}{415}\right)\frac{0.53\ldots}{25\times 36.28..} = 0.023443^2$
  - $t_{24,0.975} = 2.06\ldots$
  - $(0.428, 0.525)$

6.3 Ratio Estimation and Cluster Sampling

- Say the sociologist does not know the total population of the city.
  - How can estimate p using a ratio estimator now?
  - Estimate p under this circumstance, provide a 95% confidence interval
- The sociologist decides the rent question of her survey is more interesting and wants to pursue it further
  - She plans another study in the same city, with the same design, but this time with a 95% C.I. of width 6%
  - How many clusters should she sample?

6.4 Ratio Estimation and Two-Stage Cluster Sampling

- Recall two-stage cluster sampling
  - Once we have decided on our clusters
  - Construct a frame of all the clusters
  - Draw a SRS of clusters
  - Construct frames of all the elements in the selected clusters
  - Draw SRS of elements from each of these frames
  - The unbiased estimator is $\hat{\mu} = \left(\frac{N}{M}\right)\frac{1}{n}\sum_{i=1}^{n} M_i \bar{y}_i$

6.4 Ratio Estimation and Two-Stage Cluster Sampling

- Recall $M_i \frac{1}{m_i} \sum_{j=1}^{m_i} y_{ij} = M_i \bar{Y}_i$ is an estimator for the total value in cluster i, i.e. $M_i \bar{Y}_i = \hat{Y}_i$
  - $\Rightarrow \frac{\sum_{i=1}^{n} M_i \bar{Y}_i}{\sum_{i=1}^{n} M_i} = \frac{\sum_{i=1}^{n} \hat{Y}_i}{\sum_{i=1}^{n} M_i} := \hat{\mu}_r$ is a reasonable estimator for μ
  - It is also a ratio estimator
- Because the sampling is two stage, $\widehat{Var}(\hat{\mu}_r)$ has two parts
  - $\widehat{Var}(\hat{\mu}_r) = \frac{1}{M^2} N\,(N-n) \frac{1}{n}\, \hat{\sigma}_r^2 + \frac{1}{M^2} \frac{N}{n} \sum_{i=1}^{n} M_i (M_i - m_i) \frac{1}{m_i} \hat{\sigma}_i^2$
  - Where $\hat{\sigma}_r^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(\hat{Y}_i - \hat{\mu}_r M_i\right)^2$ and $\hat{\sigma}_i^2$ is the sample variance of Y_{ij} inside cluster i

6.4 Ratio Estimation and Two-Stage Cluster Sampling

- Compare $\widehat{Var}(\hat{\mu})$ and $\widehat{Var}(\hat{\mu}_r)$
  - The form is exactly the same, except for the term representing variance between clusters
  - The unbiased estimator uses $\hat{\sigma}_c^2$, the usual sample variance of $\hat{Y}_1, \hat{Y}_2, \dots, \hat{Y}_n$
  - The ratio estimator uses $\hat{\sigma}_r^2$, the ratio sample variance of $\hat{Y}_1, \hat{Y}_2, \dots, \hat{Y}_n$
- In this situation, the response is the estimated cluster total, $\hat{Y}_i$, and the auxiliary variable is the size of the cluster, M_i
  - If M_i and $\hat{Y}_i$ are highly correlated, we should expect the ratio estimate to perform well.

6.4 Ratio Estimation and Two-Stage Cluster Sampling

- Recall the clothes manufacturer with factories all over China
  - A clothes manufacturer has 90 factories
  - She wants to estimate the average number of hours that the sewing machines were down for repairs over the past six months
  - The plants are scattered all over China, and each factory contains many machines, so she decides to use a two-stage sampling design
  - Her resources allow her to sample $n = 10$ factories and roughly 20% of the machines in each factory
  - She knows that her company has 4500 machines in total

6.4 Ratio Estimation and Two-Stage Cluster Sampling

Factory	M_i	m_i	$\bar{y}_i$	s_i
1	50	10	5.4	3.37
2	65	13	4	3.27
3	45	9	5.666666667	4.09
4	48	10	4.8	3.65
5	52	10	4.3	3.33
6	58	12	3.833333333	3.86
7	42	8	5	2.27
8	66	13	3.846153846	2.08
9	40	8	4.875	2.47
10	56	11	5	3.44

6.4 Ratio Estimation and Two-Stage Cluster Sampling

- Plugging in the numbers…
  - $\hat{\mu}_r = \frac{\sum_{i=1}^{n} M_i \bar{Y}_i}{\sum_{i=1}^{n} M_i} = \frac{2400.179}{522} = 4.60$
  - The within variance term is the same as calculated on Slides 54 and 54 of the Cluster Sampling notes
  - Within variance term $= \frac{1}{10 \times 90 \times 50^2}(21985) = 0.0989^2$
  - $s_r^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(\hat{Y}_i - \hat{\mu}_r M_i\right)^2 = 35.13^2$
  - Between variance term $= 90 \times \frac{(90-10)}{10 \times 4500^2} \times 35.13^2 = 0.2095^2$
  - Total variance $= 0.05367 = 0.232^2$

6.4 Ratio Estimation and Two-Stage Cluster Sampling

- Thus the approximate 95% C.I. is
  - $4.60 \pm 1.96 \times 0.232 = 4.60 \pm 0.45$
- The approximate 95% C.I. using the unbiased estimator was
  - 4.80 ± 0.38
- Therefore the unbiased estimator outperforms the ratio estimator in this case
  - Why?
  - $\frac{1}{2}\frac{\hat{c}_{v,M}}{\hat{c}_{v,\hat{Y}}} \approx 0.747 > 0.544 = \hat{\rho}$, so although the correlation between M_i and $\hat{Y}_i$ was pretty good, their CVs were different enough to offset this.
- Say we do not know $M = 4500$. What could we do?

6.5 Regression Estimation

- The ratio estimator can be very useful
  - If the auxiliary variable is chosen well, the ratio estimator uses the information it provides to yield a smaller standard error
- For a ratio estimator to work well, we require the plot of response Y against auxiliary variable X to be a straight line through the origin
  - i.e. close to $Y_i = bX_i$ for some constant b
- Let's extend this a little bit
  - Consider the relationship $Y_i = a + bX_i$

6.5 Regression Estimation

- This equation is familiar from linear regression
  - Given observations $\{(x_1, y_1), \ldots, (x_n, y_n)\}$, the line of best fit through them is given by $y = a + bx$ where
    - $b = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$
    - $a = \bar{y} - b\bar{x}$
  - The value of b gives the slope of the straight line and a tells us where the line intercepts the y-axis

6.5 Regression Estimation

- Let us use this relationship to find an estimator for the mean of Y
  - $\hat{\mu}_L \coloneqq a + b\mu_x$ where
    - $b = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(X_i - \bar{X})^2}$
    - $a = \bar{Y} - b\bar{X}$
- This $\hat{\mu}_L$ is called the linear regression estimator of μ_y
- Compare it with the ratio estimator
  - $\hat{\mu}_r = \frac{\bar{Y}}{\bar{X}}\mu_x$
  - $\hat{\mu}_L = a + b\mu_x = \bar{Y} + b(\mu_x - \bar{X})$

6.5 Regression Estimation

- Much like the ratio estimator, the regression estimator is biased under SRS
  - But this bias is very small for large samples
- We may estimate the MSE of $\hat{\mu}_L$ by
  - $\widehat{Var}(\hat{\mu}_L) = \frac{N-n}{Nn}\frac{1}{n-2}\sum_{i=1}^{n}(Y_i - a - bx_i)^2$
  - where a and b are as before
- An approximate $(1-\alpha)100\%$ C.I. for μ_y is
  - $\hat{\mu}_L \pm t_{n-2,1-\frac{\alpha}{2}}\sqrt{\widehat{Var}(\hat{\mu}_L)}$
  - Notice the degrees of freedom

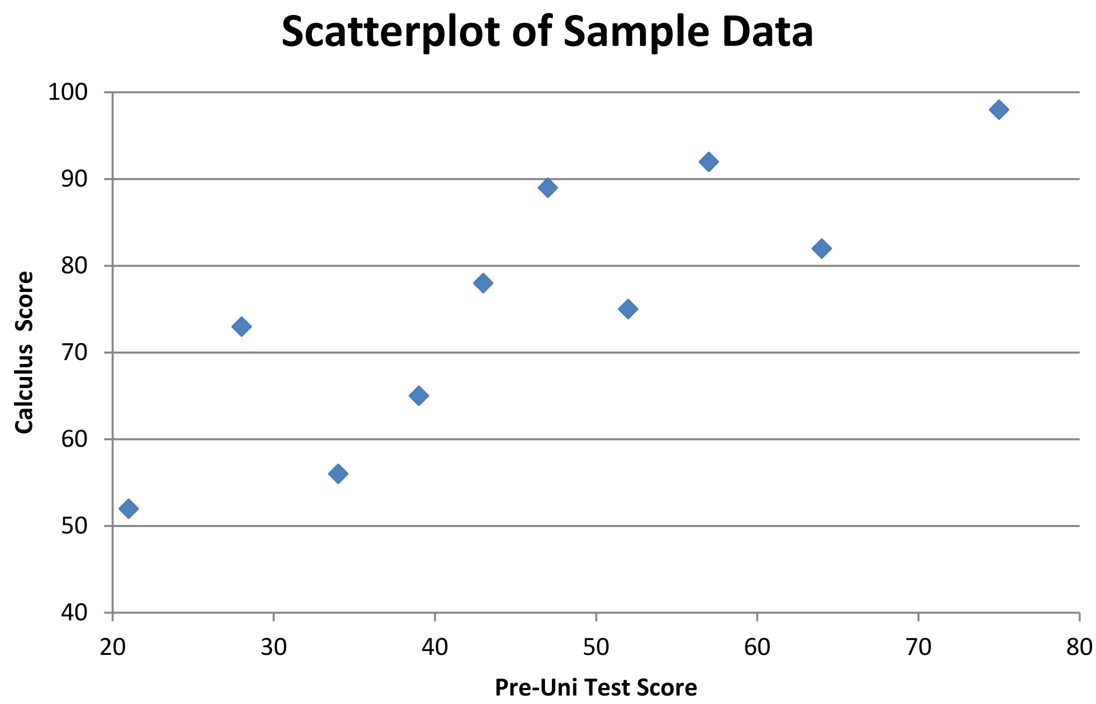
6.5 Regression Estimation

- Example (From Scheaffer et al.)
  - Before going to university, 486 students took a test. The average score was 52.
  - At university, they took a calculus course. An SRS of 10 students was taken, with the purpose of estimating the mean Calculus score of the students

Student	1	2	3	4	5	6	7	8	9	10
Pre-Uni Test Score	39	43	21	64	57	47	28	75	34	52
Calculus Score	65	78	52	82	92	89	73	98	56	75

6.5 Regression Estimation

- A quick visual check of the data suggests that there is a strong linear relationship between the response (Calculus Score) and auxiliary variable (Pre-Uni Test Score)



6.5 Regression Estimation

- Let's use the linear regression estimate, $\hat{\mu}_L$
  - $\bar{x} = 46$
  - $\bar{y} = 76$
  - $\sum_{i=1}^{10}(x_i - \bar{x})(y_i - \bar{y}) = 1894$
  - $\sum_{i=1}^{10}(x_i - \bar{x})^2 = 2474$
  - $\Rightarrow b = 0.77$
  - $a = 40.78$
  - $\Rightarrow \hat{\mu}_L = 40.78 + 0.77\mu_x = 80.59$

6.5 Regression Estimation

- Calculating the standard error
  - $\sum_{1=1}^{10}(y_i - a - bx_i)^2 = 606.03$
  - $\Rightarrow \widehat{Var}(\hat{\mu}_L) = 7.42 = 2.72^2$
  - $t_{8,0.975} = 2.31$
- Our approximate 95% C.I. for the mean calculus test score is
  - 80.59 ± 6.28
- How does this compare to the ordinary unbiased estimator?
  - $\hat{\mu} = \bar{Y}, \widehat{Var}(\hat{\mu}) = \frac{1}{N}(N - n)\frac{1}{n}\hat{\sigma}^2$

6.5 Regression Estimation

- The unbiased estimate for the mean calculus score is just the sample mean of Y
  - $\bar{Y} = ?$
  - $\hat{\sigma} = 15.11$
  - $\Rightarrow \widehat{Var}(\bar{Y}) = ?$
- Thus the 95% C.I. for μ_y using the unbiased estimate is
  - $? \pm ?$
- Which estimator performs better?

6.5 Regression Estimation

- Let's say the researcher conducting the study mistakenly uses the ratio estimator for $\hat{\mu}_y$
  - $\hat{\mu}_r = \frac{\bar{Y}}{\bar{X}}\mu_x, \widehat{Var}(\hat{\mu}_r) = \left(1 - \frac{n}{N}\right)\frac{1}{n}\hat{\sigma}_r^2$
  - Where $\hat{\sigma}_r^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(Y_i - \hat{R}X_i\right)^2$
- What C.I. does she get? Use this table

Student	1	2	3	4	5	6	7	8	9	10
$\left(y_i - \hat{R}x_i\right)^2$	0.3	48.4	299.4	563.6	4.7	128.8	715.0	671.5	0.0	119.1

6.5 Regression Estimation

- Use the right model
  - A plot of the data usually reveals which of the unbiased, ratio or linear regression estimators is appropriate
  - This visual check can be further supported by running a quick regression on the data
    - How strong is the linear relationship?
    - Is the coefficient different from zero?
  - A good choice of auxiliary variable can reduce variability of your estimate, even though it may be biased

6.5 Regression Estimation

- We have progressed from just using observations of Y to estimate population parameters of Y
  - To using auxiliary variable X when there is a proportional relationship between X and Y
  - To using auxiliary variable X when there is a linear relationship between X and Y
- What's the next step?
  - To using auxiliary variables $X_1, X_2, \dots, X_p$ when there is multiple linear regression model relationship between Y and $X_1, X_2, \dots, X_p$